

Changmin Shin

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Education

Duke University

MS IN COMPUTER SCIENCE

BS IN COMPUTER SCIENCE AND MINOR IN STATISTICAL SCIENCE

Durham, NC

Expected May 2026

Expected May 2025

- GPA: 3.77/4.0, Dean's List w/ Distinction(S'22)
- Coursework: · Advanced Software Design and Implementation · Machine Learning · Database Systems · Full-Stack IoT Systems · Advanced Web App Development · Design and Analysis of Algorithms · Computer Network · Distributed Systems

Experience

Duke University - parkinGait | Mobile Application Developer

Durham, NC

SWIFT, FIREBASE

Jun 2024 - Present

- Leading a team of 4 students under Prof. Rabih Younes to develop a mobile app for Parkinson's patients that analyzes walking exercises.
- Implementing haptic feedback to alert users when abnormal steps by utilizing real-time acceleration data to calculate step lengths.
- Collaborating with local clinics for trials, enhancing app precision, and refining features based on patient feedback for optimal performance.

Duke University - ADTG | Backend Developer

Durham, NC

JAVA, SPRING, DOCKER, KUBERNETES, POSTGRESQL

Sep 2024 - Present

- Developing an assignment delivery system for CS courses, integrating distribution, automated grading, and student progress tracking.
- Constructing a maintainable and scalable autograder while ensuring accurate evaluation of student code and grade calculation.
- Optimizing database interactions to synchronize grading results efficiently and maintain consistency across the system.

National Cancer Center | Medical Engineering Branch - SWE Intern

Remote in Seoul, Korea

TYPESCRIPT, NEXT.JS, MONGODB

Jul 2024 - Aug 2024

- Developed an online platform for sharing research and case studies on 3D printed titanium bone replacements by Dr. Hyun-Guy Kang.
- Implemented RBAC to manage content posting and facilitated patient consultations through patient data submission system.
- Built an interactive 3D map of human bones linking to case studies, with patient data submission features.

KICTECH | R&D Center - Research Engineer

Seoul, Korea

C++/ARDUINO, MICROPYTHON, C, ESP32

Jun 2023 - Aug 2023

- Developed micro-controller edge software in Arduino IDE to extract and post data from air quality sensor to Home Assistant via main server.
- Created a customized WiFi communication logic to transmit data and utilized SSDP to discover and auto-connect devices to the main server.
- Installed Internet of Things environment for apartment housing at government-hosted Land, Infrastructure and Transport Technology Fair.

Duke University | Teaching Assistant

CS371 MACHINE LEARNING, CS290 HUMAN SKILLS FOR SOFTWARE ENGINEERING

Fall 2024

- Assisted students understand machine learning concepts and led web development teams through agile programming as head TA.
- Held regular office hours, graded weekly assignments and exams, and answered questions on Ed discussion.

Projects

Menya | Web Application | TypeScript, Vue.js, MongoDB, Docker, K8s

- Developed a sample e-commerce platform specializing in ramen products that allows users to post comments and reviews.
- Constructed a CI/CD pipeline that incorporates E2E tests to ensure robust functionality before deployment.
- Leveraged Docker and Kubernetes for scalable deployment and implemented GitLab OAuth for secure user authentication.

Muckit | Foodie Mobile Application | React Native, Typescript, GraphQL

- Developed a mobile social platform that allows users to create, share, and subscribe to customizable restaurant bucket lists.
- Designed schemas for database and utilized GraphQL to effectively manage information for restaurants, users, maps, and reviews.
- Built front-end UI/UX that enables online social interaction and utilized Redux to store cache data and prevent reload from server.

Bulk Buy Buddies | Web Application | React, Typescript, Flask, MongoDB

- Developed a web application that matches Costco customers and propose bulk products to split based on users' needs.
- Utilized user input to recommend customers with similar shopping requests based on match percentage calculated from preferred location, category, and shopping time.
- Constructed a MongoDB database to archive and process shopping requests, ensuring automatic retention of only those unfulfilled.

OOGASalad(B-Cubed) | Monopoly-based Board Game | Java, JavaFX

- Developed a game building tool that allows users to create and play customized Monopoly-based board games and interact with other users.
- Built back-end and controller APIs that support construction of personalized games based on user-defined rules and data by employing dependency injection and MVC.
- Reviewed teammates' commits on weekly basis and employed test-driven development to maintain consistency and quality of code.

Skills

Language

Java, TypeScript, Python, Swift, C, C++, Go

DB

MongoDB, SQLite, GraphQL, PostgreSQL

DevOps

Git, Linux, Docker, Kubernetes, Bash

WebDev

React, Flask, Vue.js, Next.js